

Module - I

Digital Design & Binary Numbers

Number System

① Binary Number System:

0, 1
↪ 2 (Base)

② Decimal Number System:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9
↪ 10 (Base)

③ Octal Number System:

0, 1, 2, 3, 4, 5, 6, 7
↪ 8 (Base)

④ Hexadecimal Number System:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
↪ 16 (Base)

Binary Arithmetic

Binary Addition:

A	B	A+B (Carry)
0	0	0 (0)
0	1	1 (0)
1	0	1 (0)
1	1	0 (1)

$$\begin{array}{r} \overset{0}{1} \overset{0}{1} \overset{0}{0} \overset{0}{1} \overset{0}{0} \overset{0}{1} \\ 110101 \\ + 0111101 \\ \hline 10101000 \end{array}$$

Binary Subtraction:

A	B	A-B (Borrow)
0	0	0 (0)
0	1	1 (1)
1	0	1 (0)
1	1	0 (0)

$$\begin{array}{r} \overset{0}{1} \overset{2}{0} 1 \overset{0}{1} \overset{2}{0} 1 \\ - 010011 \\ \hline 011010 \end{array}$$

Binary Multiplication:

A	B	A*B
0	0	0
0	1	0
1	0	0
1	1	1

Binary Division:

A	B	A/B
0	1	0
1	1	1

Number System Conversion

1. Any Base to Decimal

(i) Binary to Decimal:

$$(1011)_2 = (?)_{10}$$

$$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= (11)_{10}$$

(ii) Octal to Decimal:

$$(35)_8 = (?)_{10}$$

$$3 \times 8^1 + 5 \times 8^0$$

$$= (29)_{10}$$

(iii) Hexa-decimal to Decimal:

$$(2A)_{16} = (?)_{10}$$

$$2 \times 16^1 + 10 \times 16^0$$

$$= (42)_{10}$$

2. Binary to Any Base

(i) Binary to Octal:

$$(110101)_2 = (?)_8$$

$$\begin{array}{ccc} \underline{110} & , & \underline{101} \\ & \searrow & \rightarrow 5 \\ & \searrow & \rightarrow 6 \end{array}$$

$$(65)_8.$$

(ii) Binary to Hexa-decimal:

$$(110111101)_2 = (?)_{16}$$

$$\begin{array}{ccc} \underline{0001} & \underline{1011} & \underline{1101} \\ 1 & 11(B) & 13(D) \end{array}$$

$$(1BD)_{16}.$$

3. Decimal to Any Base

(i) Decimal to Binary:

$$(23)_{10} = (?)_2$$

		Remainder
2	23	1
2	11	1
2	5	1
2	2	0
2	1	1
	0	

$$(10111)_2.$$

(ii) Decimal to Octal:

$$(100)_{10} = (?)_8$$

$$\begin{array}{r|l} 8 & 100 \\ 8 & 12 \\ 8 & 1 \\ & 0 \end{array}$$

Remainder

4
4
1

$$(144)_8$$

Alternatively, $\text{Decimal} \rightarrow \text{Binary} \rightarrow \text{Octal}$.

(iii) Decimal to Hexa-decimal:

$$(250)_{10} = (?)_{16}$$

$$\begin{array}{r|l} 16 & 250 \\ 16 & 15 \\ & 0 \end{array}$$

Remainder

10 (A)

15 (F)

$$(FA)_{16}$$

Alternatively, $\text{Decimal} \rightarrow \text{Binary} \rightarrow \text{Hexa-decimal}$.

4. Any Base to Binary

(i) Octal to Binary:

$$(72)_8 = (?)_2$$

$$\begin{array}{c} 7 \quad 2 \\ \swarrow \quad \searrow \\ 010 \quad 111 \end{array}$$

$$(111010)_2.$$

(ii) Hexa-decimal to Binary:

$$(C5)_{16} = (?)_2$$

$$\begin{array}{c} (12)C \quad 5 \\ \swarrow \quad \searrow \\ 1100 \quad 0101 \end{array}$$

$$(11000101)_2.$$

5. Octal to Hexa-decimal & Vice-versa

$$\text{Octal} \xleftrightarrow{3\text{-bits}} \text{Binary} \xleftrightarrow{4\text{-bits}} \text{Hexa-decimal}.$$

Conversion for Floating Number

1. Decimal to Binary

$$(26.625)_{10} = (?)_2$$

	Remainder
2 26	0 ↑
2 13	1
2 6	0
2 3	1
2 1	1
0	

$$.625 \times 2 = 1.250$$

$$.250 \times 2 = 0.500$$

$$.500 \times 2 = 1.000$$

1
0
1 ↓

$$(11010.101)_2$$

2. Binary to Decimal

$$(111.1010)_2 = (?)_{10}$$

$$1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 0 \times 2^{-4}$$

$$= 4 + 2 + 1 + \frac{1}{2} + \frac{1}{8}$$

$$= (7.625)_{10}$$

Signed Binary Numbers

1. Sign Magnitude Method [$+ \rightarrow 0$ & $- \rightarrow 1$]

(i) $(+25)_{10} = (?)_2$

	<u>Remainder</u>
2 25	1
2 12	0
2 6	0
2 3	1
2 1	1
0	

$(+25)_{10} = (011001)_2$. [Adding one 'zero' on left side.]

(ii) $(-25)_{10} = (?)_2$

	<u>Remainder</u>
2 25	1
2 12	0
2 6	0
2 3	1
2 1	1
0	

$(-25)_{10} = (111001)_2$. [Adding one '1' on left side.]

2. 1's Complement Method

1's Complement,

$$\begin{array}{l} 1 \rightarrow 0 \\ 0 \rightarrow 1 \end{array} \quad \left. \vphantom{\begin{array}{l} 1 \rightarrow 0 \\ 0 \rightarrow 1 \end{array}} \right\} \text{Replace}$$

(i) $+25 \rightarrow 011001$
 $-25 \rightarrow 100110$ } 1's complement

(ii) $+30 \rightarrow 011110$
 $-30 \rightarrow 100001$ } 1's complement

3. 2's Complement Method

2's Complement (Adding '1' to the 1's Complement).

(i) $+25 \rightarrow 011001$
 $-25 \rightarrow 100111$ } 2's complement

$+30 \rightarrow 011110$
 $-30 \rightarrow 100010$ } 2's complement

Arithmetic Operation Using 1's Complement

$$+12 \rightarrow 01100$$

$$+10 \rightarrow 01010$$

$$-12 \rightarrow 10011$$

$$-10 \rightarrow 10101$$

(i) $(+12) + (+10)$

$$\begin{array}{r} 01100 \\ + 01010 \\ \hline 10110 \end{array} \quad \times$$

Note: Always make 8-bits.

$$+12 \rightarrow 00001100$$

$$+10 \rightarrow 00001010$$

$$00010110 \rightarrow +22.$$

(ii) $+12 + (-10)$

$$+12 \rightarrow 00001100$$

$$-10 \rightarrow 11110101$$

$$00000001$$

$$\rightarrow +1$$

$$00000100 \rightarrow +1.$$

(iii) $(-12) + 10$

$$\begin{array}{r} -12 \rightarrow 11110011 \\ +10 \rightarrow \cancel{1}00001010 \\ \hline \downarrow 1111101 \end{array}$$

→ First digit is '1' so it means no. is negative.

So, calculating 1's complement of above is the result
i.e., 00000010 i.e., -2.

(iv) $(-12) + (-10)$

$$\begin{array}{r} -12 \rightarrow 11110011 \\ -10 \rightarrow \cancel{1}1110101 \\ \hline \textcircled{1}11101000 \\ \quad \quad \quad \rightarrow +1 \\ \hline \downarrow 11101001 \\ \rightarrow -ve \end{array}$$

$$= 00010110 \rightarrow -22.$$

Arithmetic Operation Using 2's Complement

(i) $(+12) + 10$

$$+12 \rightarrow 00001100$$

$$+10 \rightarrow 00001010$$

In 2's Complement,

$$\begin{array}{r} 0000\overset{\textcircled{0}}{1}100 \\ + 00001010 \\ \hline 00010110 \rightarrow +22 \end{array}$$

(ii) $+12 + (-10)$

$$+12 \rightarrow \overset{\textcircled{0}}{0}\overset{\textcircled{0}}{0}\overset{\textcircled{0}}{0}\overset{\textcircled{0}}{0}\overset{\textcircled{0}}{1}\overset{\textcircled{0}}{1}00$$

$$-10 \rightarrow +11110110$$

$$\times \boxed{1}00000010 \rightarrow +2$$

(iii) $(-12) + 10$

$$-12 \rightarrow 11110100$$

$$+10 \rightarrow +00001010$$

$$\begin{array}{r} 11111110 \\ \downarrow \\ \rightarrow -ve \end{array}$$

So, calculating the 2's complement of the final answer.

$$00000010 \rightarrow -2 \text{ (Final Answer)}$$

(iv) $(-12) + (-10)$

$-12 \rightarrow 0011 \ 1100$

$-10 \rightarrow + 1111 \ 0110$

$\begin{array}{r} 0011 \ 1100 \\ + 1111 \ 0110 \\ \hline 1010 \ 1010 \end{array} \rightarrow 1110 \ 1010$

$\rightarrow -ve$

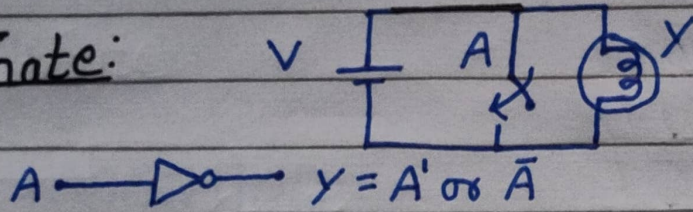
$\rightarrow -ve$

2's complement of 11101010 is 00010110
i.e. 22.

Final Answer = -22.

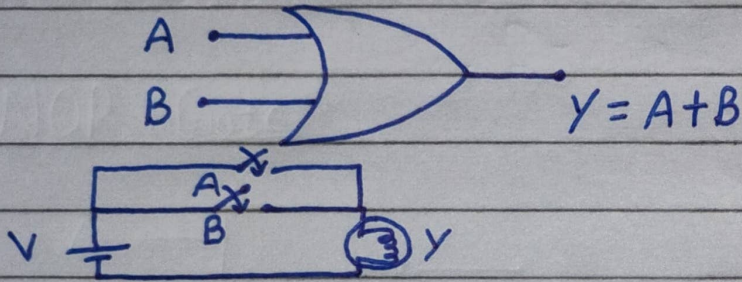
Logic Gate (Basic Gates)

NOT Gate:



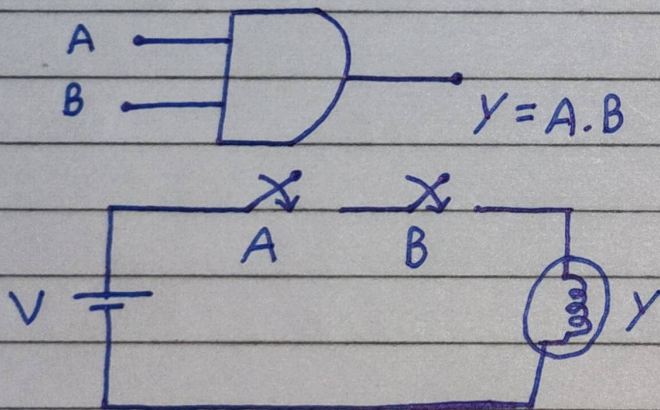
A	A'
0	1
1	0

OR Gate:



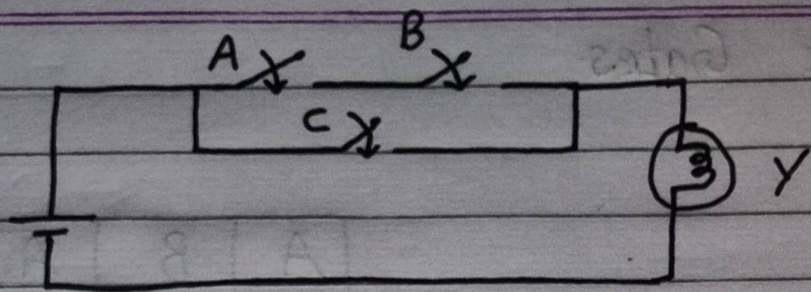
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

AND Gate:



A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

Q.

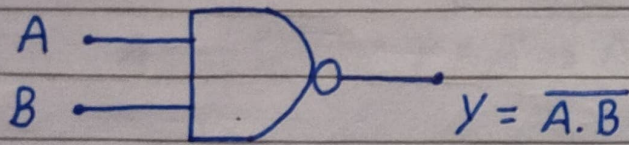


⇒

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

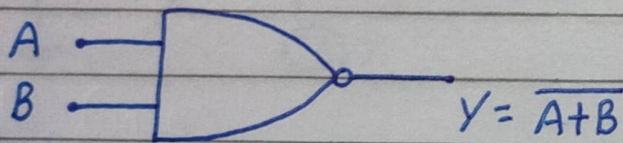
Universal Logic Gates

NAND Gate



A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

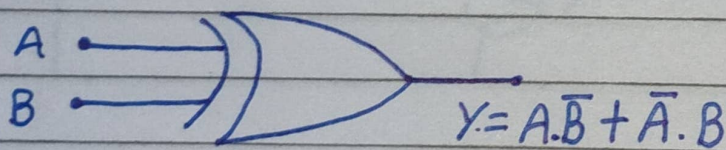
NOR Gate



A	B	$\overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

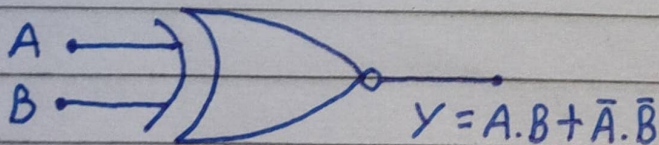
Derived Gates

X-OR Gate



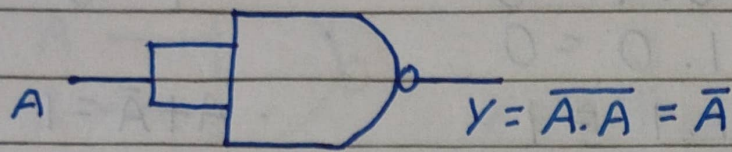
A	B	$A \oplus B$	$\overline{A \oplus B} \leftarrow A \odot B$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

X-NOR Gate

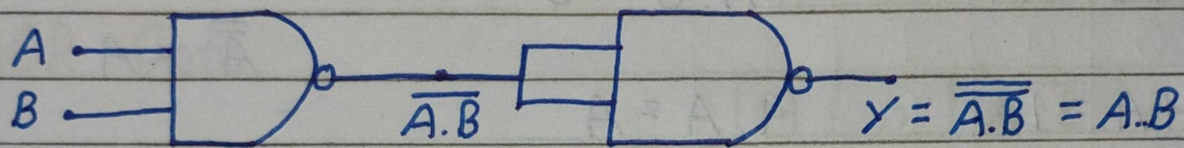


Implementation of Gate Using Universal Gate (NAND)

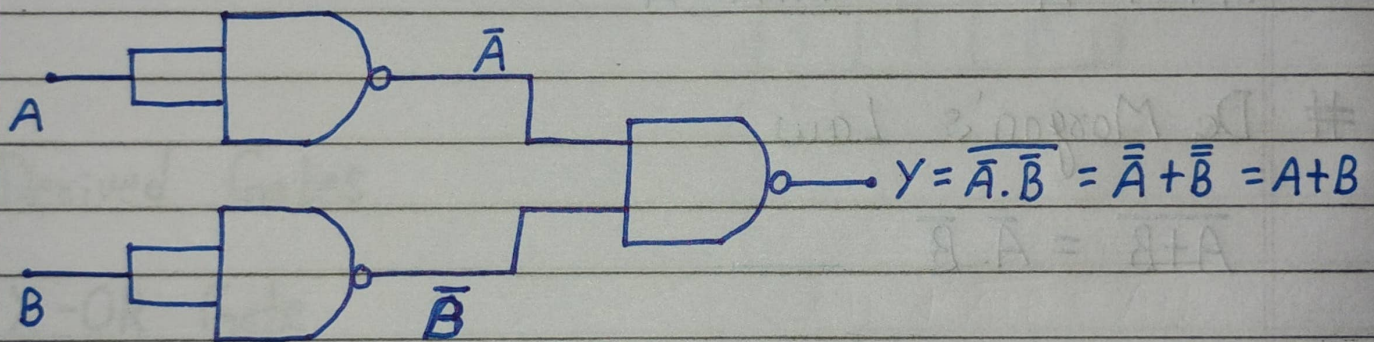
NOT Gate



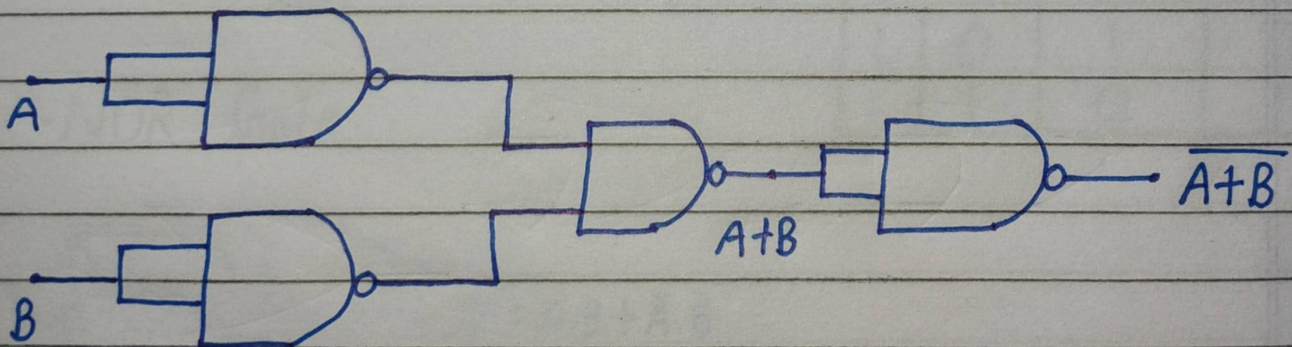
AND Gate



OR Gate

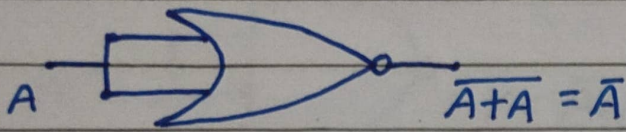


NOR Gate [OR + NOT]

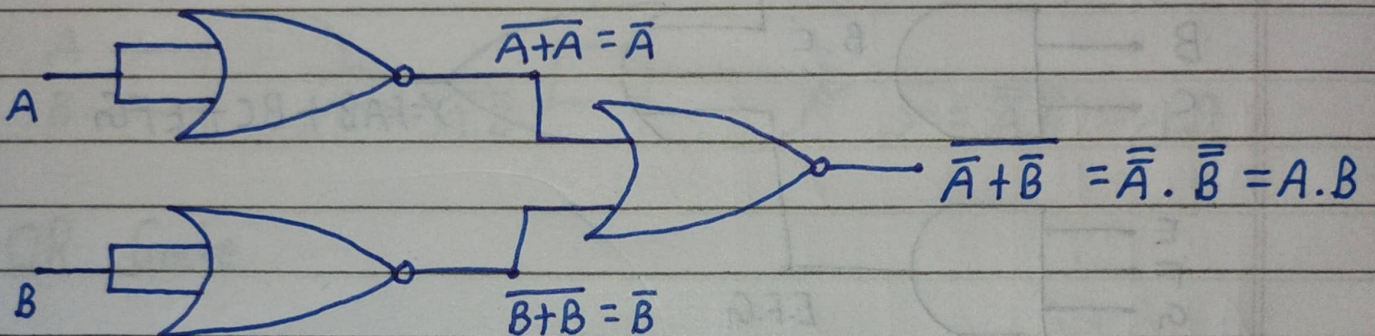


Implementation of Basic Gates Using NOR

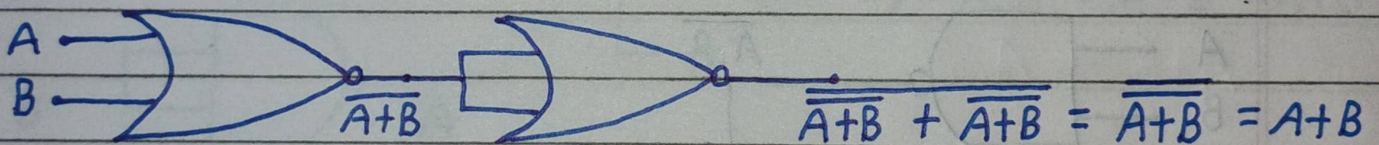
NOR Gate



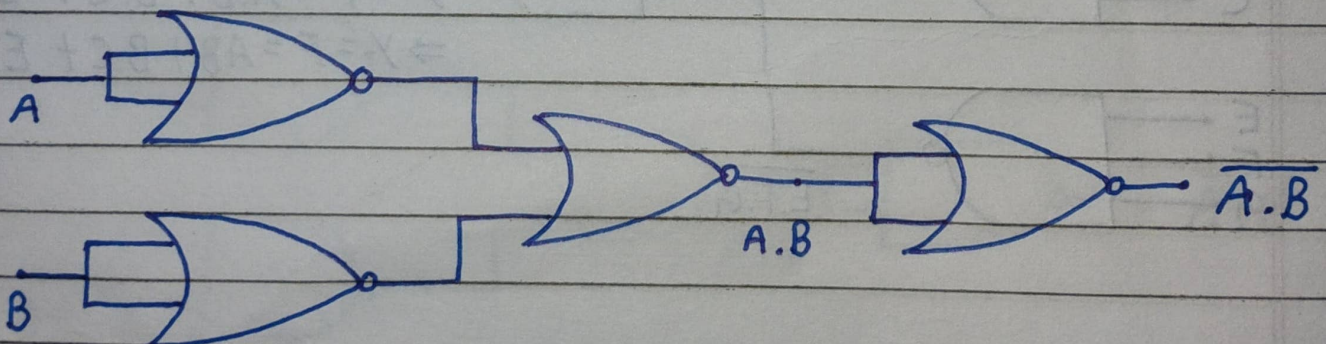
AND Gate



OR Gate



NAND Gate [AND + NOT]



Basic Rule for Boolean Algebra

$$0+0=0$$

$$0+1=1$$

$$1+0=1$$

$$1+1=1$$

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

$$\overline{0} = 1$$

$$\overline{1} = 0$$

$$A + \overline{A} = 1$$

$$0 + A = A$$

$$A + 0 = A$$

$$0 \cdot A = 0$$

$$A \cdot 0 = 0$$

$$A \cdot \overline{A} = 0$$

$$\overline{\overline{A}} = A$$

$$1 + A = 1$$

$$A + 1 = 1$$

$$1 \cdot A = A$$

$$A \cdot 1 = A$$

$$A + A = A$$

$$A \cdot A = A$$

De Morgan's Law

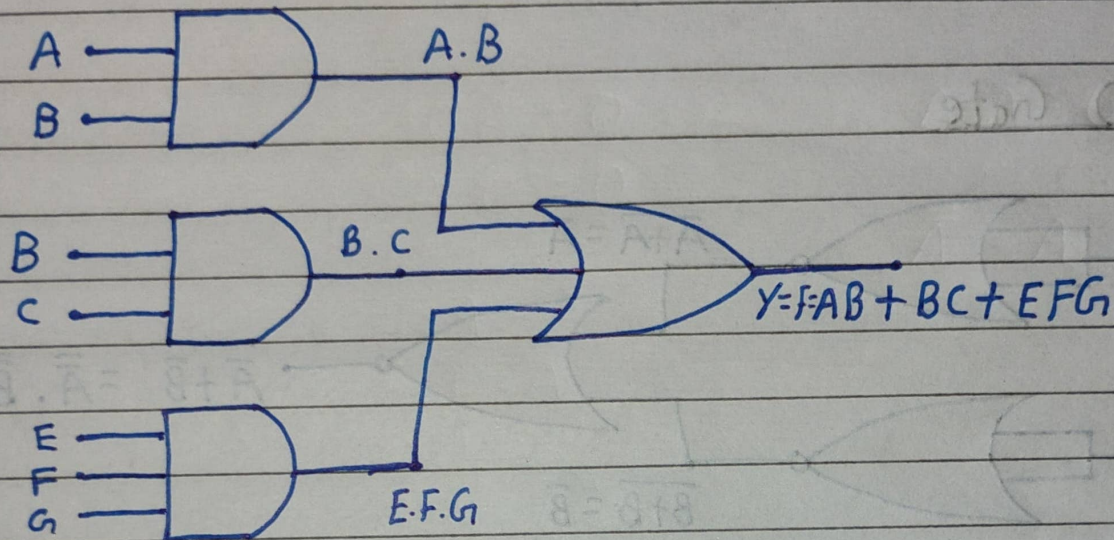
$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

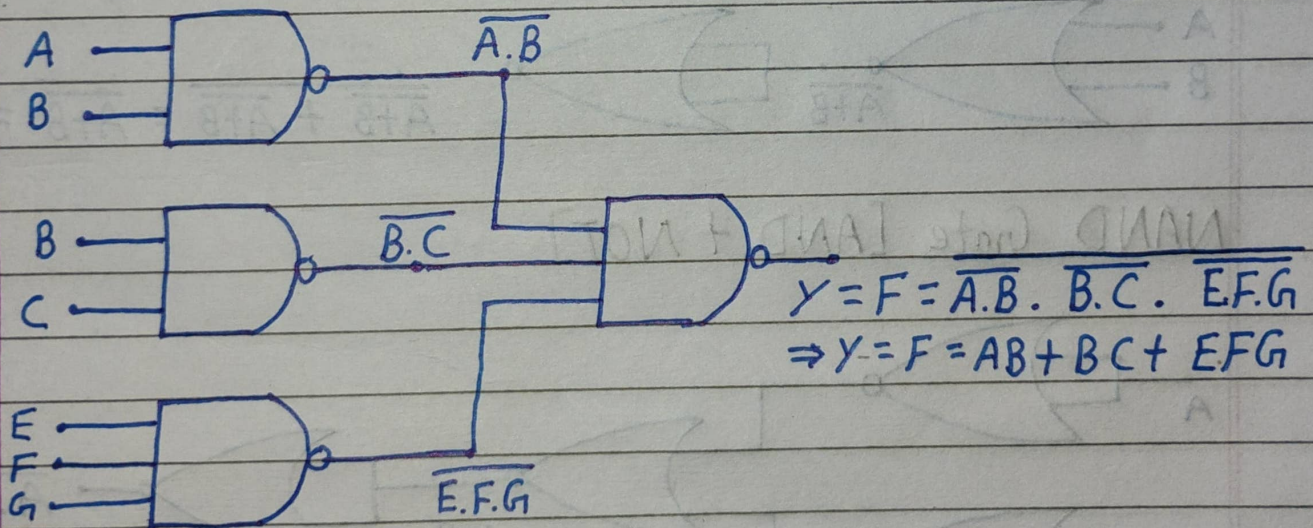
Implementation of Boolean Function using Gates

$$F = AB + BC + EFG$$

Using Basic Gates,



Using Universal (NAND) Gate,



Simplification of Boolean Function

$$F = AB + \bar{A}B + \bar{B}\bar{C} + ABC \quad [\text{Basic Method}]$$

$$F = AB(1+C) + \bar{A}B + \bar{B}\bar{C}$$

$$\Rightarrow F = AB + \bar{A}B + \bar{B}\bar{C} \quad [1+C=1]$$

$$\Rightarrow F = B(A+\bar{A}) + \bar{B}\bar{C}$$

$$\Rightarrow F = B + \bar{B}\bar{C} \quad [A+\bar{A}=1]$$

$$F = AB + \bar{A}B + \bar{B}\bar{C} + ABC \quad [\text{Standard Method}]$$

$$F = AB(C+\bar{C}) + \bar{A}B(C+\bar{C}) + \bar{B}\bar{C}(A+\bar{A}) + ABC$$

$$[x+\bar{x}=1]$$

Canonical Form

$$\Rightarrow F = ABC + AB\bar{C} + \bar{A}BC + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + ABC$$

BC A		$\bar{B}\bar{C}$		$\bar{B}C$		BC		$B\bar{C}$	
		$\bar{A}$	A	$\bar{A}$	A	$\bar{A}$	A	$\bar{A}$	A
A A	$\bar{A}$	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 0	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 1	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 3	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 2				
	A	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 4	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 5	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 7	$\overset{\circ}{A}\overset{\circ}{B}\overset{\circ}{C}$ 6				

K-Map

$$F = ABC + AB\bar{C} + \bar{A}BC + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$\Rightarrow F = \sum_m (7, 6, 3, 2, 0, 4) \quad \leftarrow \text{Standard Equation}$$